

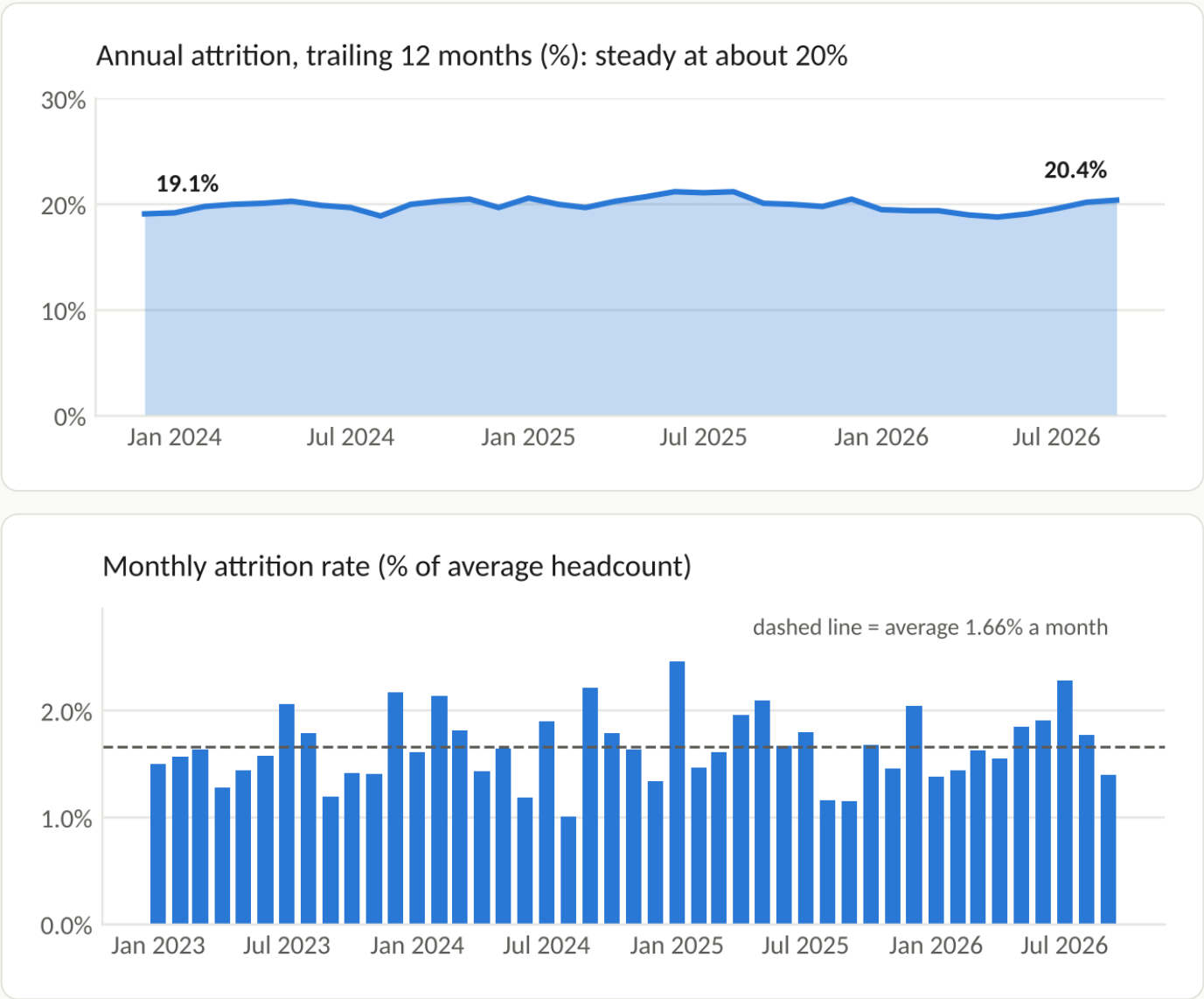
Employee Attrition Analytics

Jan 2023 - Sep 2026 • 2,600 employees • 1,035 exits • 8 departments • 25 roles

About the data: no real dataset was provided, so this analysis uses a synthetic dataset generated by `src/generate_data.py` (fixed seed, with a few deliberate data-quality problems). No names, ages, gender or contact details exist in it - only an anonymous employee ID. Figures illustrate the method, not a real company.

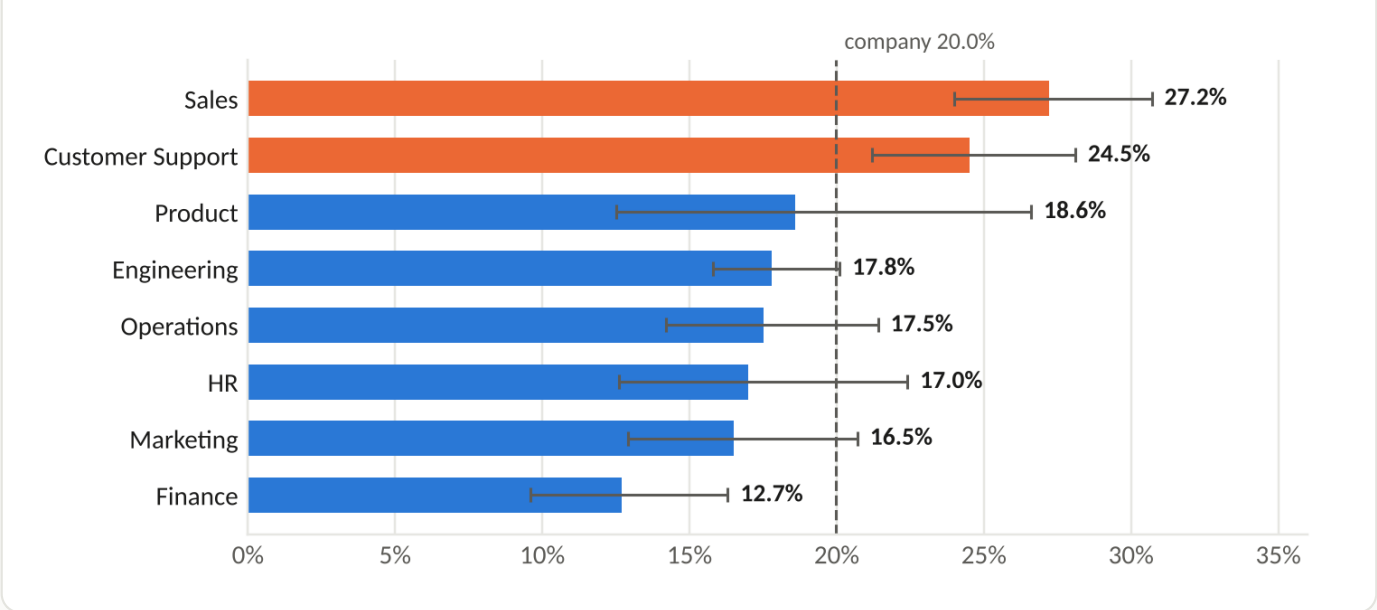
Employees analysed	Exits in window	Annual attrition	Last 12 months
2,600	1,035	20.0%	20.4%
Left within 12 months of hire	Left within 90 days of hire	Avg monthly attrition (12 mo)	Still employed
27.7%	5.6%	1.7%	1,565

1. Attrition trend

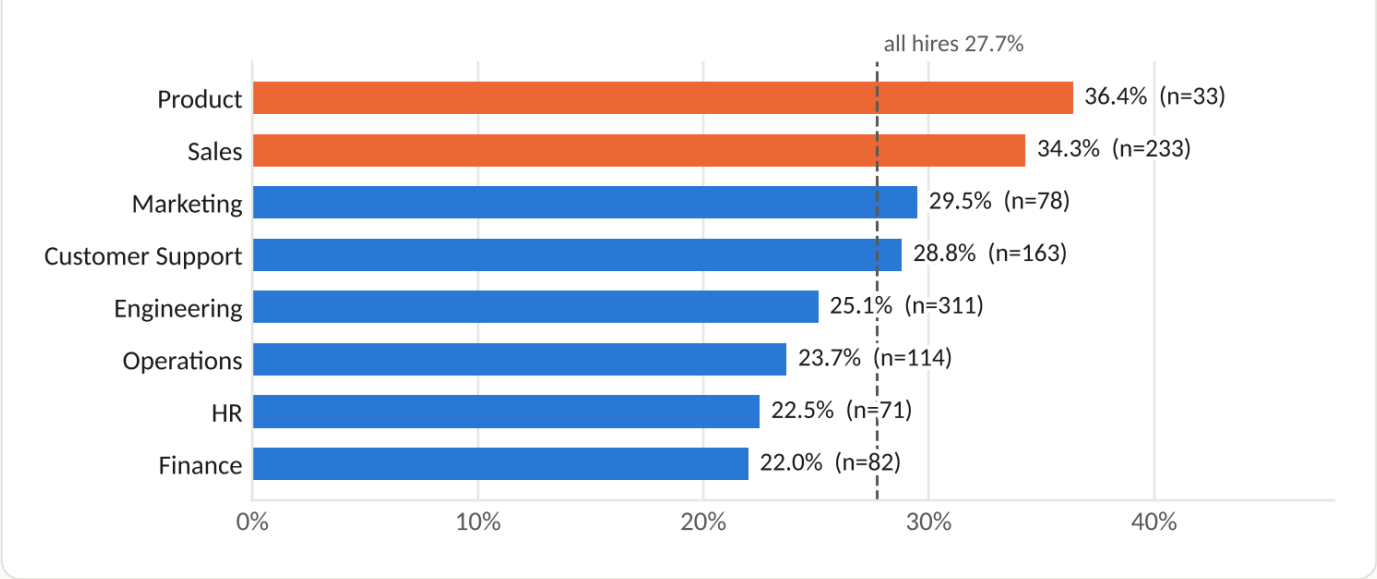


2. Departments

Sales and Customer Support lose the most people (annual attrition %, with 95% interval)

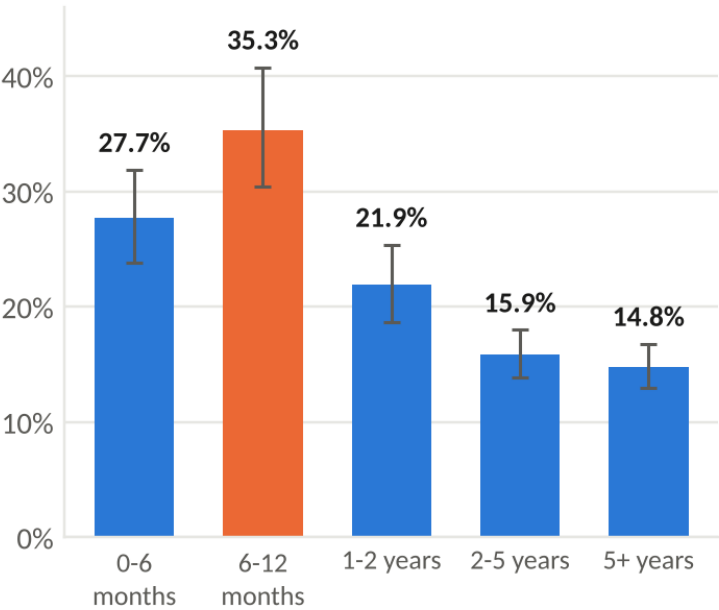


Hires who left within 12 months (%) - Product has only 33 hires, so treat it cautiously

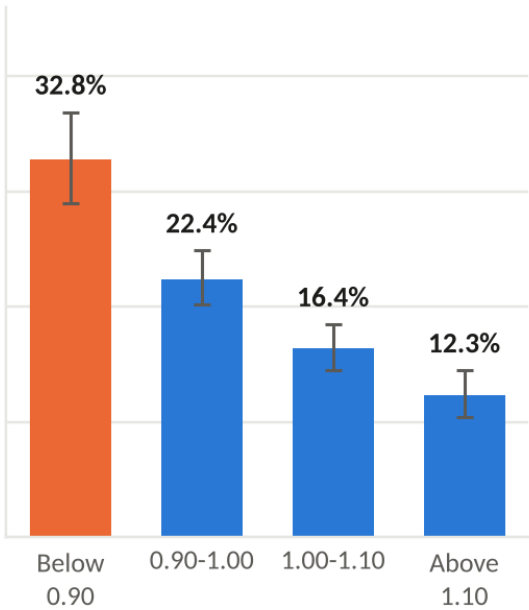


3. Tenure, pay and workload

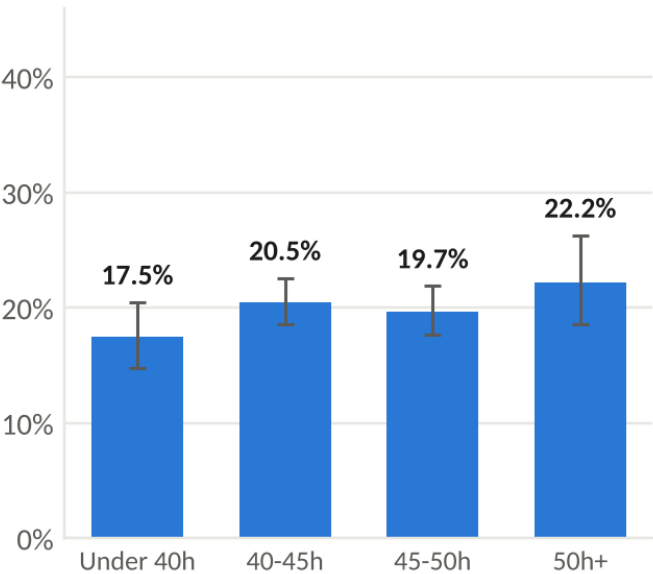
By tenure: risk peaks at 6-12 months



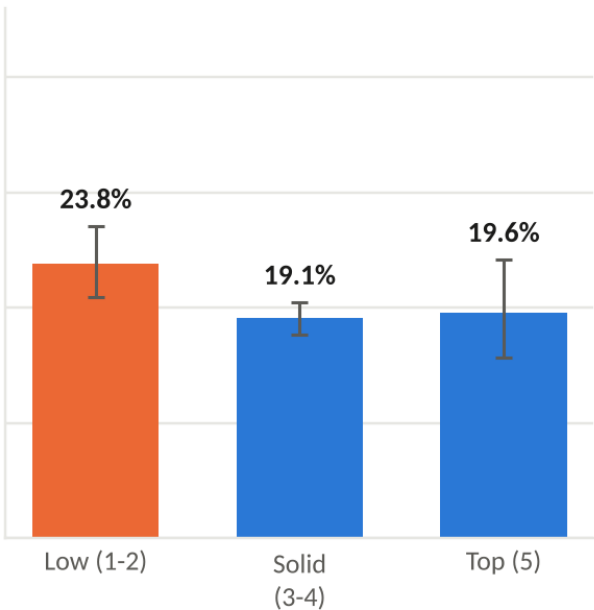
By pay position (compa-ratio)



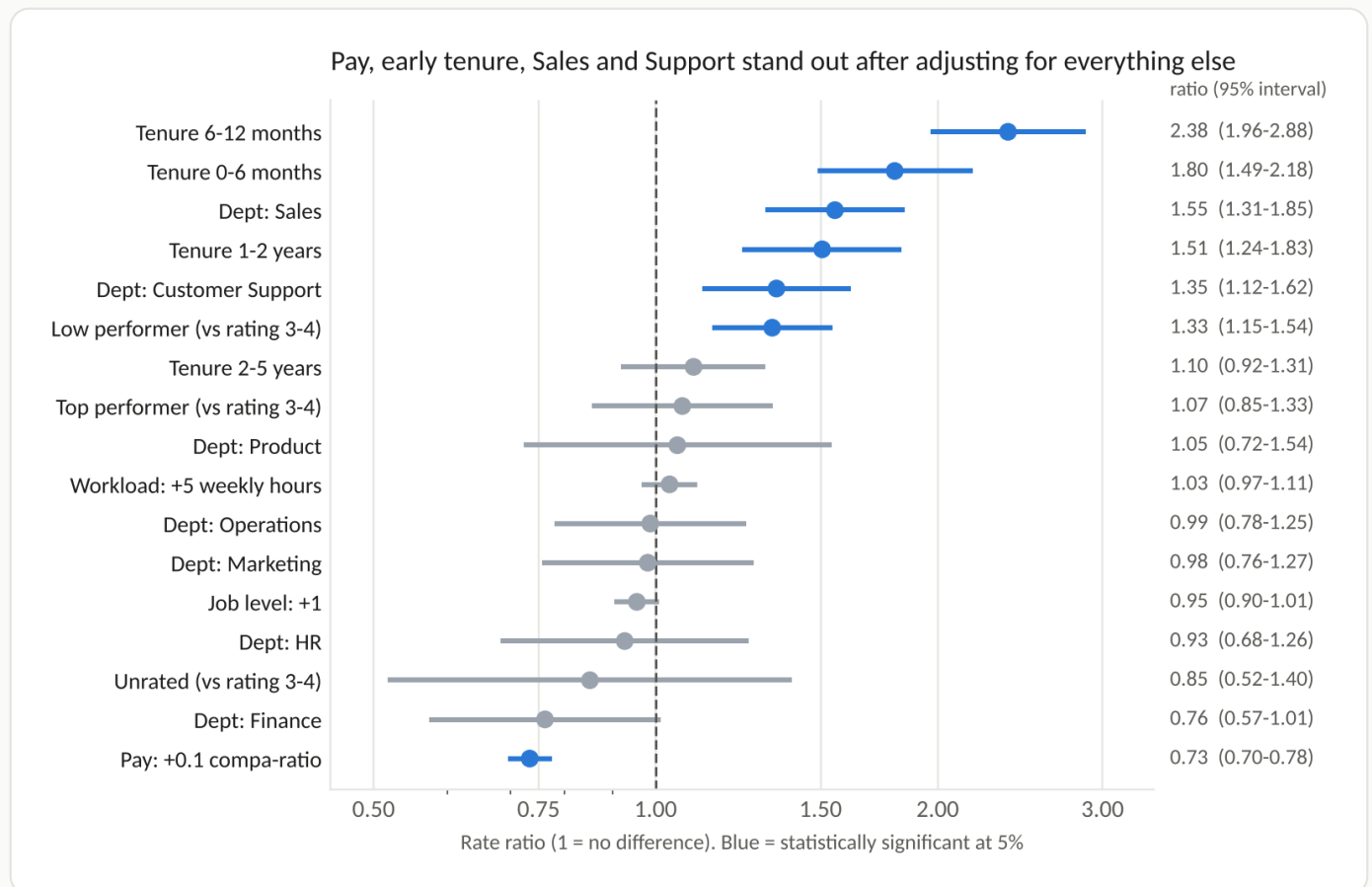
By weekly hours: weak, not significant



By performance rating



4. What drives leaving?



Rate ratios come from a Poisson regression that compares exit rates while holding the other factors equal (1.30 = a 30% higher rate; below 1 = lower). Departments are compared with Engineering, tenure bands with 5+ years, performance groups with ratings 3-4.

5. Recommendations

- Fix pay gaps first.** People paid below 0.90 of their range leave at 32.8% a year against 12.3% above 1.10. Each +0.1 of compa-ratio goes with a 27% lower exit rate. Review the 498 employees below 0.90 first, starting with Sales and Support.
- Invest in the first year.** The exit rate peaks at 6-12 months (35.3% a year, 2.4 times the 5+ year rate) and 27.7% of recent hires leave within 12 months. Add structured onboarding and manager check-ins at months 3, 6 and 9.
- Target Sales and Customer Support.** They run at 27.2% and 24.5% a year against 12.7% in Finance; adjusted, Sales is 55% and Support 35% above Engineering. Build career paths and coach managers here.
- Understand low-performer exits.** Staff rated 1-2 leave at 23.8% against 19.1% for solid performers (adjusted ratio 1.33). Check whether these are managed exits before acting.
- Watch workload, do not act yet.** Rates rise from 17.5% (under 40 hours) to 22.2% (50+), but the adjusted effect (+3.5% per extra 5 hours) is not statistically significant.
- Improve the data.** Record exit reasons (voluntary or not), dated promotions and manager IDs so the next round can separate causes from coincidence.

Method

Attrition rate = exits per 100 person-years of observed time, with exact 95% Poisson intervals; this treats people observed for different lengths of time fairly. Monthly rate = exits in the month divided by average headcount. Each person's time is split into tenure bands, so they count in a band only while they were in it. Early-tenure measures use only hires made from 2023 onward, because earlier leavers are absent from the data. Simple associations use chi-square and Mann-Whitney tests; adjusted effects come from a Poisson regression with a person-years offset, cross-checked against scikit-learn. Cleaning steps are logged below.

step	rows_affected	detail
Drop duplicate rows	5	exact repeated rows
Missing performance rating	52	kept, labelled 'Unrated'
Missing weekly hours	39	filled with the median
Exit date before hire date	0	none found
Exit flag vs exit date mismatch	0	none found
Pay or hours out of range	0	none found
Final analysis table	0	2,600 employees, 1,035 exits

Caveats

- The data is synthetic, so effect sizes reflect how it was generated, not a real workforce.
- Association is not cause: pay, tenure, department and workload overlap, and manager quality or personal reasons are not measured.
- Voluntary and involuntary exits are combined, which can blur pay-related patterns.
- Small groups (Product, HR, single roles) have wide intervals; differences there should not drive decisions.
- Promotion history is not included: a simple months-since-promotion field is tied to tenure and misleads without dated records.